SHARP BERNSTEIN–NIKOLSKII CONSTANTS FOR TRIGONOMETRIC POLYNOMIALS CONVERGE TO THOSE FOR ENTIRE FUNCTIONS WHEN $q < \infty$

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ABSTRACT. Let $V \subset \mathbb{R}^m$ be a convex body and let D_N be a differential operator with constant coefficients, homogeneous of order N. We prove that the sharp constant in the Bernstein–Nikolskii inequality $\|D_N T\|_{L^q(Q_\pi)} \le P\, a^{N+m/p-m/q} \|T\|_{L^p(Q_\pi)}$ for trigonometric polynomials T with spectrum in aV converges, as $a \to \infty$, to the sharp constant for entire functions of exponential type V in $L^p(\mathbb{R}^m)$, for every $0 < p \le q < \infty$. For $p < q$ the constant for entire functions is attained, and suitably translated near-extremal polynomials converge to an extremal function. This proves a conjecture of Ganzburg, and in one variable conjectures of Ganzburg and Tikhonov and of Levin and Lubinsky. Previously the limit was known for $q = \infty$ and in the single finite case $(p, q, N) = (2, 4, 0)$. We also show that the real and complex constants coincide when $V = -V$ and D_N has real coefficients, for every $0 < p \le q < \infty$, which answers questions of Ganzburg and of Ganzburg and Tikhonov for such operators.

1. INTRODUCTION

Let $V \subset \mathbb{R}^m$ be a convex body, that is, a compact convex set with nonempty interior. For $a > 0$ let $\mathcal{T}_{aV}$ denote the set of trigonometric polynomials

$$T(x) = \sum_{k \in aV \cap \mathbb{Z}^m} c_k e^{ik\cdot x}, \qquad c_k \in \mathbb{C},$$

and let B_V denote the set of entire functions on $\mathbb{C}^m$ of exponential type whose restrictions to $\mathbb{R}^m$ are bounded and have their distributional Fourier transform supported in V. When V is centrally symmetric, $B_V \cap L^p(\mathbb{R}^m)$ is the class of entire functions of exponential type V that lie in $L^p(\mathbb{R}^m)$, as considered by Ganzburg [2, Section 1 and Lemma 2.1]. For $0 < r < \infty$ and a measurable set $E \subseteq \mathbb{R}^m$ we write $\|F\|_{L^r(E)} = (\int_E |F|^r \,\mathrm{d}x)^{1/r}$, and we use the inequality

$$\|F_1 + F_2\|_{L^r(E)}^{\tilde r} \le \|F_1\|_{L^r(E)}^{\tilde r} + \|F_2\|_{L^r(E)}^{\tilde r}, \qquad \tilde r = \min(r, 1). \tag{1}$$

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Let $D_N = \sum_{|\alpha|=N} b_\alpha D^\alpha$, where $D^\alpha = \partial_{x_1}^{\alpha_1} \cdots \partial_{x_m}^{\alpha_m}$, be a differential operator with constant coefficients, and let $Q_\pi = \{x \in \mathbb{R}^m : |x_j| \le \pi\}$. For $0 < p \le q < \infty$, Ganzburg [2] studied the sharp constants

$$\begin{aligned} P_a &= a^{-N-m/p+m/q} \sup_{T \in \mathcal{T}_{aV} \setminus \{0\}} \frac{\|D_N T\|_{L^q(Q_\pi)}}{\|T\|_{L^p(Q_\pi)}}, \\ E &= \sup_{f \in (B_V \cap L^p(\mathbb{R}^m)) \setminus \{0\}} \frac{\|D_N f\|_{L^q(\mathbb{R}^m)}}{\|f\|_{L^p(\mathbb{R}^m)}}. \end{aligned} \tag{2}$$

In his notation these are $P_{p,q,D_N,a,V}$ and $E_{p,q,D_N,V}$. We write $P_{a,\mathbb{R}}$ and $E_{\mathbb{R}}$ for the same suprema taken over real-valued functions.

The relation between P_a and E has a long history; see the survey of Gorbachev [6]. In one variable, with $V = [-1, 1]$ and $D_N = d^s/dx^s$, the constants are $P_n = A^{(s)}_{p,q,n}$ and $E = D^{(s)}_{p,q}$ of Ganzburg and Tikhonov [4]. They proved $D^{(s)}_{p,q} \le \liminf_n A^{(s)}_{p,q,n}$ for $0 < p < q \le \infty$ [4, Theorem 1.4], proved that the limit exists and equals $D^{(s)}_{p,\infty}$ when $q = \infty$ [4, Theorem 1.5], and conjectured that the limit exists and equals $D^{(s)}_{p,q}$ for every $0 < p < q < \infty$ [4, Conjecture 1.10]. For $s = 0$ and analytic polynomials on the circle the same conjecture had been made by Levin and Lubinsky [7, (1.5)], who proved the case $q = \infty$ and identified the missing ingredient as a proof that extremal polynomials are concentrated near the point where their maximum is attained [7, Section 3]. Ganzburg extended the inequality $E \le \liminf_a P_a$ and the limit for $q = \infty$ to several variables [2, Theorem 1.3], and made the following conjecture.

Conjecture 1 (Ganzburg [2, Conjecture 1.6])**.** *Suppose that V is centrally symmetric and $0 < p \le q < \infty$. Then $\lim_{a\to\infty} P_a$ exists and equals E.*

In [3, Remark 7.7] Ganzburg explains that estimates of the form $E \ge \limsup_a P_a$ were out of reach for $q < \infty$, and the survey [6] lists the problem first among its open problems. To the best of our knowledge, the only finite case known so far is $(p, q, N) = (2, 4, 0)$ in one variable, which follows from work of Garsia, Rodemich and Rumsey [5]; see [4, (1.16)].

The main purpose of the present paper is to prove Conjecture 1, for every convex body V, and in the real-valued setting as well.

Theorem 1. *Suppose that $V \subset \mathbb{R}^m$ is a convex body and that $0 < p < q < \infty$. Then the following hold.*

1. $\lim_{a\to\infty} P_a = E < \infty$.
2. *If $E > 0$, then E is attained: there is $g \in B_V \cap L^p(\mathbb{R}^m)$ with $\|g\|_{L^p(\mathbb{R}^m)} = 1$ and $\|D_N g\|_{L^q(\mathbb{R}^m)} = E$.*
3. *Suppose that $E > 0$, that $a_j \to \infty$, and that $T_j \in \mathcal{T}_{a_j V}$ satisfy $\|T_j\|_{L^p(Q_\pi)} = a_j^{-m/p}$ and $a_j^{-N+m/q}\|D_N T_j\|_{L^q(Q_\pi)} \to E$. Then, after passing to a subsequence, there are $x_j \in \mathbb{R}^m$ and an extremal function g such that $G_j(x) = T_j((x + x_j)/a_j)$ satisfies $\|G_j - g\|_{L^p(a_j Q_\pi)} \to 0$ and $\|D_N G_j - D_N g\|_{L^q(a_j Q_\pi)} \to 0$.*

(4) *If $V = -V$, then (a), (b) and (c) hold for $P_{a,\mathbb{R}}$, $E_{\mathbb{R}}$ and real-valued functions.*

Theorem 2. *Suppose that $V \subset \mathbb{R}^m$ is a convex body and that $0 < p = q < \infty$. Then $P_a \le E$ for every $a > 0$ with $\mathcal{T}_{aV} \neq \{0\}$, and $\lim_{a\to\infty} P_a = E$. If $V = -V$, the same holds for $P_{a,\mathbb{R}}$ and $E_{\mathbb{R}}$.*

In particular, Conjecture 1 holds, together with its real version. In one variable we obtain the conjectures of Ganzburg and Tikhonov and of Levin and Lubinsky.

Corollary 1. *Suppose that $s \in \mathbb{Z}_{\ge 0}$ and $0 < p < q < \infty$. Then $\lim_n A^{(s)}_{p,q,n} = D^{(s)}_{p,q}$ and $\lim_n A^{(s)}_{p,q,n,\mathbb{R}} = D^{(s)}_{p,q,\mathbb{R}}$, and both $D^{(s)}_{p,q}$ and $D^{(s)}_{p,q,\mathbb{R}}$ are attained.*

Corollary 2. *Suppose that $0 < p < q < \infty$. Let $\lambda_{n,p,q}$ be the infimum of $\|P\|_p/\|P\|_q$ over nonzero polynomials P of degree at most $n-1$, with norms taken with respect to normalized arc length on the unit circle, and let $E_{p,q}$ be the infimum of $\|f\|_{L^p(\mathbb{R})}/\|f\|_{L^q(\mathbb{R})}$ over nonzero entire functions f of exponential type at most π in $L^p(\mathbb{R})$. Then $n^{1/p-1/q}\lambda_{n,p,q} \to E_{p,q}$, and $E_{p,q}$ is attained.*

We also settle a side question of Ganzburg and Tikhonov, whether the real and complex constants coincide. The equalities $A^{(s)}_{p,q,n} = A^{(s)}_{p,q,n,\mathbb{R}}$ and $D^{(s)}_{p,q} = D^{(s)}_{p,q,\mathbb{R}}$ hold for $q = \infty$ [4, Theorem 1.1], and for $p = q$, where all four constants equal 1 [4, Section 1]; they state that the remaining case $p < q < \infty$ is open [4, Section 1]. Ganzburg asks the same question in several variables [2, Remark 1.7].

Proposition 1. *Suppose that $V \subset \mathbb{R}^m$ is a convex body with $V = -V$, that the coefficients b_α of D_N are real, and that $0 < p \le q < \infty$. Then $P_a = P_{a,\mathbb{R}}$ for every $a > 0$, and $E = E_{\mathbb{R}}$.*

Corollary 3. *Suppose that $s \in \mathbb{Z}_{\ge 0}$, $n \ge 1$ and $0 < p \le q < \infty$. Then $A^{(s)}_{p,q,n} = A^{(s)}_{p,q,n,\mathbb{R}}$ and $D^{(s)}_{p,q} = D^{(s)}_{p,q,\mathbb{R}}$.*

The proof of Proposition 1 is an averaging argument, a variant of one used by Marcinkiewicz and Zygmund [9] for operators from L^p to L^q with $p \le q$. The only points to check are that it applies to the classes $\mathcal{T}_{aV}$ and B_V, and that it does not use the triangle inequality in L^p, so that $p < 1$ is allowed.

Let us briefly describe our approach. Put $G(x) = T(x/a)$, so that P_a is the best constant in $\|D_N G\|_{L^q(aQ_\pi)} \le P_a \|G\|_{L^p(aQ_\pi)}$ for $2\pi a$-periodic functions of exponential type V. Take near-extremal functions G_j along a sequence realizing $\limsup_a P_a$, normalized by $\|G_j\|_{L^p} = 1$ and translated so that $|G_j(0)|$ is bounded below. They converge locally uniformly to some $g \in B_V$ with $0 < \lambda = \|g\|_p^p \le 1$. If $\lambda < 1$, part of the mass of G_j escapes to infinity. The Brezis–Lieb lemma splits both norms exactly into the part carried by g and the part carried by $G_j - g$, and $G_j - g$ is, up to a small error, again a periodic function of exponential type V, namely G_j minus a Levitan-type

polynomial built from g. Applying the definition of P_a to this remainder gives

$$\limsup_a P_a^q \le E^q \lambda^{q/p} + \bigl(\limsup_a P_a\bigr)^q (1-\lambda)^{q/p}.$$

Since $\lambda^{q/p} + (1-\lambda)^{q/p} \le 1$ when $p < q$, this forces $\limsup_a P_a \le E$. In contrast to the proofs for $q = \infty$ in [2, 4, 7], which work at the point where the maximum is attained, we never show that near-extremal polynomials are concentrated: a near-extremal sequence may split, but splitting cannot increase the ratio when $p < q$. This is the concentration-compactness principle of Lions [8]. For $p = q$ the splitting gives no information, and we use instead a direct transference from periodic functions to functions in $L^p(\mathbb{R}^m)$.

Organization. The present paper is organized into six additional sections. In Section 2 we collect the Nikolskii and Bernstein inequalities that we need, with constants independent of a, including the range $r < 1$. In Section 3 we construct the Levitan-type polynomials and prove the lower bound $E \le \liminf_a P_a$ in the present generality. Section 4 contains the proof of Theorem 1, and Section 5 the proof of Theorem 2. In Section 6 we prove Proposition 1. In Section 7 we deduce the corollaries, and Section 8 contains some concluding remarks.

2. Preliminaries

Throughout, we fix a point ξ_0 in the interior of V and $\rho > 0$ such that the ball of radius ρ about ξ_0 lies in V; when $V = -V$ we take $\xi_0 = 0$. The Fourier transform is $\hat f(\xi) = \int f(x) e^{-i\xi\cdot x}\,\mathrm{d}x$, extended to tempered distributions; the normalization plays no role. We write $C_a = aQ_\pi$ and

$$\mathcal{G}_a = \{G(x) = T(x/a) : T \in \mathcal{T}_{aV}\},$$

the $2\pi a$-periodic elements of B_V. For $T \in \mathcal{T}_{aV}$ and $G(x) = T(x/a)$ we have $D_N G(x) = a^{-N}(D_N T)(x/a)$ and $\|G\|_{L^r(C_a)} = a^{m/r}\|T\|_{L^r(Q_\pi)}$, so that

$$P_a = \sup_{G \in \mathcal{G}_a \setminus \{0\}} \frac{\|D_N G\|_{L^q(C_a)}}{\|G\|_{L^p(C_a)}}. \tag{3}$$

From now on we work with (3).

We fix a real even function φ in the Schwartz class with $\hat\varphi = 1$ on a neighbourhood of $V \cup (-V)$. Then $f = f * \varphi$ and $D_N f = f * D_N\varphi$ for every $f \in B_V$, and in particular for every $G \in \mathcal{G}_a$. So D_N acts on these functions as convolution with the Schwartz function $k = D_N\varphi$. This is the only property of D_N that we use; see Section 8.

To control D_N in L^r for $r < 1$ we use Peetre's maximal function

$$f_M^*(x) = \sup_{y \in \mathbb{R}^m} \frac{|f(x-y)|}{(1+|y|)^M}, \qquad M > 0,$$

and the Hardy–Littlewood maximal operator $\mathcal{M}$.

Lemma 1. *Suppose that $0 < r < \infty$. There is a constant C, independent of $a \geq 1$, such that the following hold.*

1. *(Nikolskii) $\|G\|_\infty \leq C\|G\|_{L^r(C_a)}$ for $G \in \mathcal{G}_a$, and $\|f\|_\infty \leq C\|f\|_{L^r(\mathbb{R}^m)}$ for $f \in B_V$.*
2. *(Bernstein in L^∞) $\|\nabla f\|_\infty \leq C\|f\|_\infty$ for $f \in B_V$.*
3. *(Peetre) If $M > m/r$, then $\|G_M^*\|_{L^r(C_a)} \leq C\|G\|_{L^r(C_a)}$ for $G \in \mathcal{G}_a$, and $\|f_M^*\|_{L^r(\mathbb{R}^m)} \leq C\|f\|_{L^r(\mathbb{R}^m)}$ for $f \in B_V$.*
4. *(Bernstein in L^r) We have $\|D_N G\|_{L^r(C_a)} \leq C\|G\|_{L^r(C_a)}$ for $G \in \mathcal{G}_a$, and $\|D_N f\|_{L^r(\mathbb{R}^m)} \leq C\|f\|_{L^r(\mathbb{R}^m)}$ for $f \in B_V$. Consequently, if $0 < p \leq r \leq q < \infty$, then $\|D_N G\|_{L^q(C_a)} \leq C\|G\|_{L^r(C_a)}$ and $\|D_N f\|_{L^q(\mathbb{R}^m)} \leq C\|f\|_{L^r(\mathbb{R}^m)}$. In particular, $\sup_{a\geq 1} P_a < \infty$ and $E < \infty$.*

Proof. (a) Since $|\varphi(y)| \leq C(1+|y|)^{-m-1}$, the periodized function $\Phi_a(y) = \sum_{j\in\mathbb{Z}^m} |\varphi(y + 2\pi a j)|$ satisfies $\Phi_a \leq C_0$ for all $a \geq 1$, and $\int_{C_a} \Phi_a = \|\varphi\|_1$. Let $\theta = \min(r, 1)$. Since $G = G * \varphi$,

$$|G(x)| \leq \int_{\mathbb{R}^m} |G(y)||\varphi(x-y)|\,\mathrm{d}y \leq \|G\|_\infty^{1-\theta} \int_{C_a} |G(y)|^\theta \Phi_a(x-y)\,\mathrm{d}y.$$

If $r \leq 1$, the integral is at most $C_0\|G\|_{L^r(C_a)}^r$, and we find that $\|G\|_\infty^r \leq C_0\|G\|_{L^r(C_a)}^r$. If $r > 1$, the claim follows from Hölder's inequality and the bound

$$\|\Phi_a\|_{L^{r'}(C_a)} \leq C_0^{1/r}\|\varphi\|_1^{1/r'}.$$

For $f \in B_V$ we replace Φ_a by $|\varphi|$.

(b) This follows from $\nabla f = f * \nabla\varphi$.

(c) Let $\theta = m/M < r$. Peetre's inequality [11, Theorem 1.3.1] states that $f_M^* \leq C(\mathcal{M}|f|^\theta)^{1/\theta}$ pointwise for every bounded f with $\operatorname{supp}\hat{f} \subset V$. On $\mathbb{R}^m$ the claim follows from the maximal theorem in $L^{r/\theta}$, since $r/\theta > 1$. For $G \in \mathcal{G}_a$ we need the periodic maximal bound $\|\mathcal{M}u\|_{L^s(C_a)} \leq C\|u\|_{L^s(C_a)}$ for $2\pi a$-periodic $u \geq 0$ and $s > 1$, with C independent of $a \geq 1$. Let $x \in C_a$. Balls of radius at most πa about x lie in $3C_a$, so their averages are bounded by $\mathcal{M}(u\mathbf{1}_{3C_a})(x)$. A ball of radius $t > \pi a$ meets at most $(3t/(\pi a))^m$ translates of C_a, so its average is at most $C|C_a|^{-1}\int_{C_a} u \leq C(|C_a|^{-1}\int_{C_a} u^s)^{1/s}$. Taking $L^s(C_a)$ norms and using the maximal theorem for $u\mathbf{1}_{3C_a}$ proves the bound.

(d) We have $|D_N f(x)| \leq \int |k(y)||f(x-y)|\,\mathrm{d}y \leq f_M^*(x)\int |k(y)|(1+|y|)^M\,\mathrm{d}y$, so the first two bounds follow from (c). For the consequence, combine the first bound at exponent q with $\|G\|_{L^q(C_a)}^q \leq \|G\|_\infty^{q-r}\|G\|_{L^r(C_a)}^r$ and (a). □

For $r \geq 1$, part (d) is the classical Bernstein inequality; for $r < 1$ in one variable it is due to Arestov, and the maximal function argument gives it in the generality needed here, with constants independent of a.

The splitting in Section 4 rests on the following lemma, which is valid for every exponent in $(0, \infty)$.

Lemma 2 (Brezis–Lieb [1])**.** *Suppose that $0 < r < \infty$, that $\sup_j \|F_j\|_{L^r(\mathbb{R}^m)} < \infty$, and that $F_j \to F$ almost everywhere. Then $\|F_j\|_{L^r}^r - \|F_j - F\|_{L^r}^r \to \|F\|_{L^r}^r$.*

In words: if F_j converges pointwise to F, then the r-th power of the norm of F_j splits exactly into the contribution of F and the contribution of the remainder $F_j - F$, no matter where the remainder lives.

We also need an averaging property of periodic functions.

Lemma 3. *Suppose that $u \in L^\infty(\mathbb{R}^m)$ is periodic with respect to a lattice $b\mathbb{Z}^m$, let $\langle u\rangle$ be its mean over a period, and let $\psi \in L^1(\mathbb{R}^m)$. Then $L^{-m}\int u(x)\psi(x/L)\,\mathrm{d}x \to \langle u\rangle \int \psi$ as $L \to \infty$.*

Proof. The left side equals $\int u(Ly)\psi(y)\,\mathrm{d}y$, and both sides are bounded by $\|u\|_\infty\|\psi\|_1$. By density it is therefore enough to take for ψ the indicator function of a box Q. Then LQ contains $L^m|Q|b^{-m} + O(L^{m-1})$ whole periods, and the remainder has volume $O(L^{m-1})$. □

3. Levitan-type polynomials and the lower bound

The remainder in the splitting must again be a polynomial. To arrange this we replace a function $h \in B_V \cap L^p$ by a periodic function of exponential type V that agrees with h up to a small error, in the manner of Levitan's polynomials [2, 4]. We first shrink the spectrum of h slightly, so that there is room to multiply by a wide window.

Let $\kappa(u) = (\sin u/u)^2$ and $\bar\kappa(u) = \min(1, u^{-2})$. Then $\kappa \le \bar\kappa$, and since $1 + |u| \le (1 + |u - v|)(1 + |v|)$ and $(1 + |u|)^{-2} \le \bar\kappa(u) \le 4(1 + |u|)^{-2}$,

$$\bar\kappa(u - v) \le 4(1 + |v|)^2\bar\kappa(u). \tag{4}$$

Put $K_a(x) = \prod_{l=1}^m \kappa(x_l/(2a))$ and $\bar K_a(x) = \prod_l \bar\kappa(x_l/(2a))$. Since $\hat\kappa$ is supported in $[-2, 2]$, $\hat K_a$ is supported in $[-1/a, 1/a]^m$. For functions w with $\sum_j |w(x+2\pi aj)| < \infty$ write $\mathrm{Per}_a\, w(x) = \sum_{j\in\mathbb{Z}^m} w(x+2\pi aj)$, and for bounded h put $S_a h = \mathrm{Per}_a(hK_a)$. For $0 < \tau < 1$ let $V_\tau = \xi_0 + \tau(V - \xi_0)$. By convexity, every point within distance $(1 - \tau)\rho$ of V_τ lies in V.

The next lemma says that periodizing a function dominated by $\bar K_a H$, with $H \in L^r$, changes it only by a small amount in $L^r(C_a)$.

Lemma 4. *Suppose that $0 < r < \infty$, that $H \in L^r(\mathbb{R}^m)$ is nonnegative, that $|w_a| \le \bar K_a H$, and that $w_a \to w$ locally uniformly as $a \to \infty$. Then $\|\mathrm{Per}_a\, w_a - w\|_{L^r(C_a)} \to 0$.*

Proof. Let $x \in C_a$ and $j \in \mathbb{Z}^m$. Since $|x_l + 2\pi a j_l| \ge \pi a|j_l|$, we have $\bar K_a(x + 2\pi aj) \le \omega_j = \prod_l \omega(j_l)$, where $\omega(0) = 1$ and $\omega(i) = 4/(\pi^2 i^2)$ for $i \ne 0$. These weights have sum $W = \sum_j \omega_j = (7/3)^m$. By subadditivity of $t \mapsto t^r$ when $r \le 1$, and by Jensen's inequality with weights ω_j/W when $r > 1$, we get for every $J \subseteq \mathbb{Z}^m$

$$\Big|\sum_{j\in J} w_a(x + 2\pi aj)\Big|^r \le W^{\max(r-1,0)} \sum_{j\in J} H(x + 2\pi aj)^r.$$

Fix $M > 0$, let $a > M$ and put $Q_M = [-M, M]^m$. On Q_M we have $|\mathrm{Per}_a\, w_a - w| \le |w_a - w| + |\sum_{j\ne 0} w_a(\cdot + 2\pi aj)|$. The first term tends to 0 uniformly. For the second, the sets $Q_M + 2\pi aj$ with $j \ne 0$ are disjoint and

lie outside C_a, so its r-th power integrates to at most $C\int_{\mathbb{R}^m\setminus C_a} H^r$, which tends to 0. On $C_a \setminus Q_M$ the sets $(C_a \setminus Q_M) + 2\pi aj$ are disjoint and avoid Q_M; since also $|w| \le H$, we find that

$$\int_{C_a\setminus Q_M} |\mathrm{Per}_a\, w_a|^r + \int_{C_a\setminus Q_M} |w|^r \le C\int_{\mathbb{R}^m\setminus Q_M} H^r.$$

Hence $\limsup_a \|\mathrm{Per}_a\, w_a - w\|_{L^r(C_a)}^{\tilde r} \le C(\int_{\mathbb{R}^m\setminus Q_M} H^r)^{\tilde r/r}$ by (1), and we let $M \to \infty$. □

Lemma 5. *Suppose that $0 < \tau < 1$ and that $h \in B_V$ with $\operatorname{supp}\hat h \subset V_\tau$. Then the following hold.*

1. *$S_a h \in \mathcal{G}_a$ whenever $a \ge \sqrt{m}/((1-\tau)\rho)$.*
2. *If $h \in L^p(\mathbb{R}^m)$ for some $0 < p < \infty$, then for every $r \ge p$, $\|S_a h - h\|_{L^r(C_a)} \to 0$ and $\|D_N S_a h - D_N h\|_{L^r(C_a)} \to 0$ as $a \to \infty$.*
3. *If h is real-valued, then so is $S_a h$.*

Proof. (a) The function hK_a is integrable, and its spectrum lies in $V_\tau + [-1/a, 1/a]^m \subset V$. The series defining $\mathrm{Per}_a(hK_a)$ converges uniformly on compact sets, since $\sum_j \bar K_a(x + 2\pi aj) \le W$ on C_a with uniformly small tails. Its Fourier coefficients are $(2\pi a)^{-m}\widehat{hK_a}(k/a)$, which vanish unless $k \in aV$. A continuous periodic function with finitely many nonzero Fourier coefficients is the corresponding trigonometric polynomial, so $S_a h \in \mathcal{G}_a$.

(b) We apply Lemma 4 twice. For the first limit take $w_a = hK_a$, $w = h$ and $H = |h|$; here $H \in L^r$ because $h \in L^p \cap L^\infty$. For the second take $w_a = D_N(hK_a)$ and $w = D_N h$. By Fubini's theorem, $\mathrm{Per}_a\, D_N(hK_a) = D_N S_a h$, and $|D_N(hK_a) - D_N h| \le \int |k(y)||h(x-y)||K_a(x-y) - 1|\,\mathrm{d}y \to 0$ locally uniformly. By (4), $\bar K_a(x-y) \le 4^m(1+|y|)^{2m}\bar K_a(x)$ for $a \ge 1$, and so

$$|D_N(hK_a)(x)| \le 4^m \bar K_a(x)\int |k(y)|(1+|y|)^{2m}|h(x-y)|\,\mathrm{d}y \le C\bar K_a(x)h_M^*(x)$$

with $M > m/r$. By Lemma 1 (c), $H = Ch_M^* \in L^r$.

(c) This holds because K_a is real. □

For $g \in B_V$ and $0 < \tau < 1$ we put $g_\tau(x) = e^{i(1-\tau)\xi_0\cdot x} g(\tau x)$. Then $\operatorname{supp}\widehat{g_\tau} \subset V_\tau$, $|g_\tau(x)| = |g(\tau x)|$, and $g_\tau \to g$ locally uniformly as $\tau \to 1$. If $g \in L^p$, then also $\|g_\tau - g\|_p \to 0$, by continuity of dilations in L^p and dominated convergence. When $V = -V$ we have $\xi_0 = 0$, so $g_\tau = g(\tau\,\cdot)$ is real if g is.

With these polynomials, the lower bound of Ganzburg and Tikhonov [4, Theorem 1.4] and Ganzburg [2, Theorem 1.3] extends to the present setting. We include the short proof.

Proposition 2. *Suppose that $0 < p \le q < \infty$. Then $E \le \liminf_{a\to\infty} P_a$, and if $V = -V$, then $E_{\mathbb{R}} \le \liminf_{a\to\infty} P_{a,\mathbb{R}}$.*

Proof. Let $f \in B_V \cap L^p$ be nonzero, and let $0 < \tau < 1$. By Lemma 5, $S_a f_\tau \in \mathcal{G}_a$ for large a, and

$$\liminf_{a\to\infty} P_a \geq \lim_{a\to\infty} \frac{\|D_N S_a f_\tau\|_{L^q(C_a)}}{\|S_a f_\tau\|_{L^p(C_a)}} = \frac{\|D_N f_\tau\|_q}{\|f_\tau\|_p}.$$

As $\tau \to 1$, $D_N f_\tau \to D_N f$ pointwise by dominated convergence, and $\|f_\tau\|_p = \tau^{-m/p}\|f\|_p$. Fatou's lemma gives $\|D_N f\|_q \leq \liminf_a P_a \|f\|_p$. The real case is identical. □

4. Proof of Theorem 1

By Lemma 1 (d), $\beta = \limsup_{a\to\infty} P_a$ is finite. In view of Proposition 2 it is enough to prove that $\beta \leq E$, and we may assume that $\beta > 0$.

Near-extremal functions do not vanish. Choose $a_j \to \infty$ with $P_{a_j} \to \beta$, and $G_j \in \mathcal{G}_{a_j}$ with $\|G_j\|_{L^p(C_{a_j})} = 1$ and $\|D_N G_j\|_{L^q(C_{a_j})} \to \beta$. Fix $r \in (p, q)$. By Lemma 1 (d), for large j,

$$\beta/2 \leq \|D_N G_j\|_{L^q(C_{a_j})} \leq C\|G_j\|_{L^r(C_{a_j})} \leq C\|G_j\|_\infty^{1-p/r}.$$

Hence $\|G_j\|_\infty \geq 2\eta$ for some $\eta > 0$. Since $\mathcal{G}_a$ is invariant under translations, we may translate each G_j so that $|G_j(0)| \geq \eta$.

A limit function. By Lemma 1 (a) and (b), the functions G_j and their gradients are uniformly bounded. By the Arzelà–Ascoli theorem, a subsequence converges locally uniformly to a bounded continuous function g with $|g(0)| \geq \eta$. Since $G_j \to g$ as tempered distributions and $\operatorname{supp} \hat{G}_j \subset V$, we have $g \in B_V$. By dominated convergence, $D_N G_j = G_j * k \to D_N g$ locally uniformly. Fatou's lemma gives $0 < \lambda = \|g\|_p^p \leq 1$, and so $\|D_N g\|_q \leq E\lambda^{1/p}$.

Splitting. We apply Lemma 2 to $\mathbf{1}_{C_{a_j}} G_j \to g$ in L^p and to $\mathbf{1}_{C_{a_j}} D_N G_j \to D_N g$ in L^q. Since also $\|g\|_{L^p(\mathbb{R}^m\setminus C_a)} \to 0$ and $\|D_N g\|_{L^q(\mathbb{R}^m\setminus C_a)} \to 0$, we obtain

$$\|G_j - g\|^p_{L^p(C_{a_j})} \to 1 - \lambda, \qquad \|D_N G_j - D_N g\|^q_{L^q(C_{a_j})} \to \beta^q - \|D_N g\|_q^q. \tag{5}$$

The remainder $G_j - g$ is not periodic, so we replace g by a Levitan-type polynomial. Fix $0 < \tau < 1$ and put $R_j = G_j - S_{a_j} g_\tau$. By Lemma 5 (a), $R_j \in \mathcal{G}_{a_j}$ for large j, and by (3),

$$\|D_N R_j\|_{L^q(C_{a_j})} \leq P_{a_j}\|R_j\|_{L^p(C_{a_j})}.$$

Let $\delta = \|g_\tau - g\|_p$, which tends to 0 as $\tau \to 1$. By Lemma 5 (b) and (1), $\limsup_j \|S_{a_j} g_\tau - g\|^{\tilde{p}}_{L^p(C_{a_j})} \leq \delta^{\tilde{p}}$. Since $g_\tau - g \in B_V \cap L^p$, we have $\|D_N(g_\tau - g)\|_q \leq E\delta$, and hence $\limsup_j \|D_N S_{a_j} g_\tau - D_N g\|^{\tilde{q}}_{L^q(C_{a_j})} \leq (E\delta)^{\tilde{q}}$. Combining these bounds with (5) and (1), we find that

$$\big(\beta^q - \|D_N g\|_q^q\big)^{\tilde{q}/q} - (E\delta)^{\tilde{q}} \leq \liminf_j \|D_N R_j\|^{\tilde{q}}_{L^q(C_{a_j})} \leq \beta^{\tilde{q}}\big((1-\lambda)^{\tilde{p}/p} + \delta^{\tilde{p}}\big)^{\tilde{q}/\tilde{p}}.$$

The function g does not depend on τ. Letting $\tau \to 1$ and writing $\varrho = q/p > 1$, we obtain

$$\beta^q \le \|D_N g\|_q^q + \beta^q(1-\lambda)^\varrho \le E^q\lambda^\varrho + \beta^q(1-\lambda)^\varrho. \tag{6}$$

Conclusion. For $0 < \lambda \le 1$ we have $1-(1-\lambda)^\varrho \ge \lambda^\varrho$, so (6) gives $\beta^q\lambda^\varrho \le E^q\lambda^\varrho$. Since $\lambda > 0$, $\beta \le E$. This proves part (a).

For parts (b) and (c), suppose that $E > 0$ and run the same argument for a sequence as in (c), with G_j as defined there; after the translation above we may take x_j as the translation used. By part (a) the ratios tend to $E = \lim_j P_{a_j}$, so the steps above apply with $\beta = E$. Now (6) reads $E^q(1-(1-\lambda)^\varrho) \le \|D_N g\|_q^q \le E^q\lambda^\varrho$. Since $\lambda^\varrho + (1-\lambda)^\varrho < 1$ for $0 < \lambda < 1$, this forces $\lambda = 1$ and $\|D_N g\|_q = E$. So g is extremal, and by (5) both differences tend to zero. Applied to a maximizing sequence, this proves (b).

For part (d), take every G_j real-valued. Then g and $g_\tau = g(\tau\,\cdot)$ are real, and so is $S_a g_\tau$ by Lemma 5 (c). Hence R_j is real and $\|D_N R_j\| \le P_{a_j,\mathbb{R}}\|R_j\|$, and Proposition 2 gives the lower bound. □

The strict inequality $\lambda^\varrho + (1-\lambda)^\varrho < 1$ for $0 < \lambda < 1$ is where the assumption $p < q$ enters. For $q = \infty$ Lemma 2 is not available, and that case is covered by [4, Theorem 1.5] and [2, Theorem 1.3]. For $p = q$ the first step fails: the exponentials $e^{ik\cdot x/a}$ spread their mass evenly over C_a, and E need not be attained. For instance, for $p = q = 2$ and $D_N = \partial/\partial x_1$, Plancherel's theorem shows that $E = \max_{\xi\in V}|\xi_1|$, which is not attained in L^2.

5. Proof of Theorem 2

In view of Proposition 2 it is enough to prove that $P_a \le E$. The idea is to transfer a periodic function to $L^p(\mathbb{R}^m)$ by multiplying it with a wide window, which costs nothing when $p = q$. Fix a and a nonzero $G \in \mathcal{G}_a$, $G(x) = \sum_k c_k e^{ik\cdot x/a}$. For $0 < \tau < 1$ put $G_\tau(x) = e^{i(1-\tau)\xi_0\cdot x}G(\tau x)$, which has spectrum in V_τ. Then

$$D_N G_\tau(x) = e^{i(1-\tau)\xi_0\cdot x}\sum_k c_k\,\hat{k}\big(\tau k/a + (1-\tau)\xi_0\big)e^{i\tau k\cdot x/a},$$

so $|G_\tau|$ and $|D_N G_\tau|$ are bounded and $2\pi a/\tau$-periodic. Their period means satisfy $\langle|G_\tau|^p\rangle = \langle|G|^p\rangle$ and $\langle|D_N G_\tau|^p\rangle \to \langle|D_N G|^p\rangle$ as $\tau \to 1$, the latter because the sum is finite and $\hat{k}$ is continuous.

Choose $K \in \mathbb{N}$ with $2Kp > 1$, let $\vartheta(x) = \prod_l(\sin x_l/x_l)^{2K}$ and $\vartheta_L = \vartheta(\cdot/L)$. Then $\hat{\vartheta}$ is supported in $[-2K, 2K]^m$, and $|\vartheta|$ and $|\nabla\vartheta|$ are at most $C\Psi$, where $\Psi(x) = \prod_l(1+|x_l|)^{-2K}$ satisfies $\Psi(z-w) \le \prod_l(1+|w_l|)^{2K}\Psi(z)$. For large L the function $F_L = G_\tau\vartheta_L$ belongs to $B_V \cap L^p$, since its spectrum lies in $V_\tau + [-2K/L, 2K/L]^m \subset V$. By Lemma 3,

$$L^{-m}\|F_L\|_p^p \to \langle|G|^p\rangle\|\vartheta\|_p^p, \qquad L^{-m}\|(D_N G_\tau)\vartheta_L\|_p^p \to \langle|D_N G_\tau|^p\rangle\|\vartheta\|_p^p.$$

It remains to show that the commutator $c_L = D_N F_L - (D_N G_\tau)\vartheta_L$ is negligible. We have $c_L(x) = \int k(y)G_\tau(x-y)(\vartheta_L(x-y) - \vartheta_L(x))\,\mathrm{d}y$. By the mean value

theorem and the property of Ψ, for $L \geq 1$,

$$|c_L(x)| \leq \frac{C}{L}\Psi(x/L)U(x), \qquad U(x) = \int |k(y)||y| \prod_l (1+|y_l|)^{2K} |G_\tau(x-y)| \,\mathrm{d}y.$$

Here U is bounded and periodic, so Lemma 3 gives $L^{-m}\|c_L\|_p^p = O(L^{-p})$. By (1),

$$E \geq \lim_{L\to\infty} \frac{\|D_N F_L\|_p}{\|F_L\|_p} = \frac{\langle |D_N G_\tau|^p \rangle^{1/p}}{\langle |G|^p \rangle^{1/p}} \longrightarrow \frac{\|D_N G\|_{L^p(C_a)}}{\|G\|_{L^p(C_a)}} \qquad (\tau \to 1).$$

If $V = -V$ and G is real, then $\xi_0 = 0$ and F_L is real. □

For $p < q$ the same windowing gives only $\|D_N F_L\|_q / \|F_L\|_p \approx L^{m/q - m/p} \to 0$: spreading a periodic function over many periods costs a factor which tends to zero exactly when $p < q$. This is why the two cases require different arguments.

6. Real and complex constants

We first record the averaging argument in an abstract form. For $0 < r < \infty$ put $c_r = (2\pi)^{-1} \int_0^{2\pi} |\cos\theta|^r \,\mathrm{d}\theta$.

Lemma 6. *Suppose that $0 < p \leq q < \infty$, that (Ω, λ) and (Ω', λ') are σ-finite measure spaces, that X is a complex vector space of measurable functions on Ω with $\operatorname{Re}(e^{i\theta}F) \in X$ for every $F \in X$ and $\theta \in \mathbb{R}$, and that D is a linear map from X to measurable functions on Ω' with $D\operatorname{Re}(e^{i\theta}F) = \operatorname{Re}(e^{i\theta}DF)$. If $\|DF\|_{L^q(\lambda')} \leq C\|F\|_{L^p(\lambda)}$ for every real-valued $F \in X$, then the same inequality holds for every $F \in X$.*

Proof. Let $F \in X$ and put $F_\theta = \operatorname{Re}(e^{i\theta}F)$. Since $|\operatorname{Re}(e^{i\theta}w)| = |w||\cos(\theta + \arg w)|$, Fubini's theorem gives

$$\frac{1}{2\pi}\int_0^{2\pi} \|DF_\theta\|_{L^q(\lambda')}^q \,\mathrm{d}\theta = c_q \|DF\|_{L^q(\lambda')}^q.$$

Applying the hypothesis to each real-valued F_θ, we find that this is at most $C^q(2\pi)^{-1}\int_0^{2\pi}\|F_\theta\|_{L^p(\lambda)}^q \,\mathrm{d}\theta$. Write $\|F_\theta\|_{L^p(\lambda)}^p = \int g_\theta \,\mathrm{d}\lambda$ with $g_\theta(x) = |F(x)|^p|\cos(\theta + \arg F(x))|^p$. Since $q/p \geq 1$, Minkowski's integral inequality in $L^{q/p}(\mathrm{d}\theta/2\pi)$ gives

$$\Big(\frac{1}{2\pi}\int_0^{2\pi}\Big(\int g_\theta \,\mathrm{d}\lambda\Big)^{q/p}\mathrm{d}\theta\Big)^{p/q} \leq \int\Big(\frac{1}{2\pi}\int_0^{2\pi} g_\theta^{q/p}\,\mathrm{d}\theta\Big)^{p/q}\mathrm{d}\lambda = c_q^{p/q}\|F\|_{L^p(\lambda)}^p.$$

Hence $c_q\|DF\|_{L^q(\lambda')}^q \leq C^q c_q \|F\|_{L^p(\lambda)}^q$, and $c_q > 0$. □

Proof of Proposition 1. Restriction to real-valued functions gives $P_{a,\mathbb{R}} \leq P_a$ and $E_\mathbb{R} \leq E$. For the reverse inequalities we apply Lemma 6 with $D = D_N$.

For P_a, take $X = \mathcal{T}_{aV}$ on Q_π. If $T = \sum c_k e^{ik\cdot x}$, then $\overline{T} = \sum \overline{c_k} e^{-ik\cdot x}$ has spectrum in $-aV = aV$, so $\operatorname{Re}(e^{i\theta}T) = (e^{i\theta}T + e^{-i\theta}\overline{T})/2 \in \mathcal{T}_{aV}$. Since the b_α are real, $D_N\overline{T} = \overline{D_N T}$, and hence $D_N \operatorname{Re}(e^{i\theta}T) = \operatorname{Re}(e^{i\theta}D_N T)$. Applying the lemma with $C = P_{a,\mathbb{R}}a^{N+m/p-m/q}$ gives $P_a \leq P_{a,\mathbb{R}}$.

For E, take $X = B_V \cap L^p(\mathbb{R}^m)$, restricted to $\mathbb{R}^m$. For $f \in B_V$ let $f^*(z) = \overline{f(\overline{z})}$. It is entire of exponential type, its restriction to $\mathbb{R}^m$ is $\overline{f}$, which is bounded and has spectrum in $-V = V$, so $f^* \in B_V$. Hence $(e^{i\theta}f + e^{-i\theta}f^*)/2$ lies in B_V, and its restriction to $\mathbb{R}^m$ is $\operatorname{Re}(e^{i\theta}f)$, which lies in $L^p(\mathbb{R}^m)$ since $|\operatorname{Re}(e^{i\theta}f)| \le |f|$. As before $D_N \operatorname{Re}(e^{i\theta}f) = \operatorname{Re}(e^{i\theta}D_N f)$ on $\mathbb{R}^m$, and the lemma with $C = E_{\mathbb{R}}$ gives $E \le E_{\mathbb{R}}$. □

The lemma also holds for $q = \infty$, with the supremum over θ in place of the average; this recovers [4, Theorem 1.1].

7. Proofs of the corollaries

Proof of Corollary 1. Take $m = 1$, $V = [-1, 1]$, $D_N = d^s/dx^s$ and $a = n$. Then $\mathcal{T}_{nV}$ consists of the trigonometric polynomials of degree at most n, and (2) gives $P_n = A^{(s)}_{p,q,n}$. Every entire function of exponential type at most 1 in $L^p(\mathbb{R})$ is bounded on $\mathbb{R}$ and has spectrum in $[-1, 1]$ [2,10], and conversely. Hence $E = D^{(s)}_{p,q}$, and likewise in the real case. Now apply Theorem 1. □

Proof of Corollary 3. With $m = 1$, $V = [-1, 1]$, $D_N = d^s/dx^s$ and $a = n$ as in the proof of Corollary 1, this is Proposition 1. □

Proof of Corollary 2. Take $m = 1$, $V = [0, 1]$, $\xi_0 = 1/2$, $N = 0$ and $a = n-1$. Then $aV \cap \mathbb{Z} = \{0, \dots, n-1\}$, so $G(x) = P(e^{ix/a})$ runs over $\mathcal{G}_{n-1}$ as P runs over the polynomials of degree at most $n-1$. Since $\int_{C_a} |G|^r = 2\pi a \|P\|_r^r$, we have $1/\lambda_{n,p,q} = (2\pi a)^{1/p-1/q} P_{n-1}$. The map $h \mapsto f(x) = e^{-i\pi x} h(2\pi x)$ is a bijection from $B_{[0,1]} \cap L^p$ onto the entire functions of exponential type at most π in $L^p(\mathbb{R})$, and $\|f\|_r = (2\pi)^{-1/r}\|h\|_r$. Hence $1/E_{p,q} = (2\pi)^{1/p-1/q} E$, and Theorem 1 gives

$$n^{1/p-1/q}\lambda_{n,p,q} = \Big(\frac{n}{2\pi(n-1)}\Big)^{1/p-1/q} \frac{1}{P_{n-1}} \to \frac{(2\pi)^{-(1/p-1/q)}}{E} = E_{p,q}.$$

The infimum $E_{p,q}$ is attained because E is. □

8. Concluding remarks

Fourier multipliers. The only property of D_N used above is that it acts on B_V and on $\mathcal{G}_a$ as convolution with a kernel k satisfying $\int |k(y)|(1+|y|)^M \,\mathrm{d}y < \infty$ for a suitable M; one may take $M > 2m + m/p$ for Theorem 1 and $M > \max(2m + m/p, 1 + 2Km)$ for Theorem 2. Hence both theorems hold, with the same proofs, for every Fourier multiplier whose symbol is smooth on a neighbourhood of V, for instance finite differences, and Riesz or Weyl derivatives when $0 \notin V$ or the order is an even integer. For the multiplier $|\xi|^\alpha$ with $0 \in V$ the kernel decays only like $|x|^{-m-\alpha}$, and small fractional orders are not covered.

Problem 1. *Do Theorems 1 and 2 hold for the fractional derivative with symbol $|\xi|^\alpha$, $0 < \alpha < 1$, when $0 \in V$?*

Complex coefficients. Proposition 1 also holds when the coefficients b_α are a common unimodular multiple of real numbers, since multiplying D_N by a unimodular constant changes neither constant. For general complex coefficients the hypothesis cannot simply be dropped: a numerical experiment with $m = 2$, $V = [-1, 1]^2$, $a = 1$, $D_N = \partial_1 + i\partial_2$, $p = 2$ and $q = 4$ gives $P_a \approx 0.69$ and $P_{a,\mathbb{R}} \approx 0.62$. We have not examined whether E and $E_\mathbb{R}$ can differ for such operators, so the question of Ganzburg [2, Remark 1.7] remains open for complex coefficients.

Monotonicity. Ganzburg and Tikhonov also conjectured that $A^{(s)}_{p,q,n}$ decreases in n [4, Conjecture 1.3]. Our argument says nothing about this: it compares the constants only in the limit.

Problem 2 (Ganzburg and Tikhonov)**.** *Is $A^{(s)}_{p,q,n}$ a nonincreasing function of n for $0 < p < q < \infty$?*

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